COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 7 : SLO – 1 | BCD representation of decimal numbers
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Fill in the Blanks (5)

• 1. 4
• 2. 0001 0010
• 3. Binary Coded Decimal
• 4. 1001 1001
• 5. display (digital display devices like calculators and digital clocks)

Match the Following

• A. 9 — 4. 1001
• B. 15 — 1. 0001 0101
• C. 28 — 5. 0010 1000
• D. 76 — 3. 0111 0110
• E. 100 — 2. 0001 0000 0000

Activity Task: Convert and Create

• Example birthdate: 15-08 (DDMM) → digits 1, 5, 0, 8.
• BCD: 1 = 0001, 5 = 0101, 0 = 0000, 8 = 1000.
• Birthdate in BCD: 0001 0101 0000 1000.
• (Each decimal digit is independently encoded into its 4-bit 8421 binary code — this is what makes BCD ideal for digital date/time displays.)